

2018 International Biology Olympiad

National Team Selection Camp

Theoretical assessment questions

Part One

Student Code



Precautions:

1. This section of the exam consists of 12 pages (including the cover) and is worth 100 points. Please check if you have it.
If any pages are missing, please inform the test administrator immediately.
2. When answering the questions, please check that the cover of the test paper matches your test number.
3. Please answer the questions on the test paper using a blue (black) pen, and please keep the test paper clean.
4. You may write your answers on the back of the questions, but please indicate the question number.

Part 1: (100 points)

I. International Competition-Style Questions (35 points in total)

Note: International competition questions are graded at unequal intervals. Each question has four parts; a score of 0 is given for all four parts answered incorrectly, and a score of 0 is given for each part answered correctly (parts 1, 2, 3, and 4).

They received 0.5, 1, 2.5 , and 5 points respectively .

1. Cancer cells are difficult to prevent and treat because they can evade the immune system's surveillance. (Stanford University, Stem Cell Biology and Regeneration)

A research team at the Institute for Stem Cell Biology and Regenerative Medicine pointed out that...

Cancer cells overexpress CD47, sending a "don't eat me" message to macrophages through the CD47 pathway, helping them evade detection.

Attack by innate immunity. They confirmed that CD47 interacts with signaling regulatory proteins on the surface of macrophages.

It binds to γ (signal regulatory protein γ , SIRP γ), inhibiting its toxic effect on cancer cells. They later discovered that...

The two "don't eat me" message transmission pathways are via γ 2-microglobulin (γ 2M) and macrophages.

Cancer cells evade macrophage attack by binding to leukocyte immunoglobulin-like receptor B1 (LILRB1) on the cell surface. γ 2M is a type I major histocompatibility complex.

MHC-1 (a component of the class I major histocompatibility complex) is highly expressed on cancer cells.

Low MHC-1 levels can affect the efficacy of anti-CD47 antibodies; the higher the MHC-1 expression, the more resistant the cancer cells are to anti-CD47 antibodies.

High. Blocking MHC-1 or LILRB1 can enhance the phagocytic activity of macrophages against cancer cells. Additionally, certain cancer cells...

Cells can also evade T cell toxicity by reducing the expression of MHC-1 on the cell membrane.

Determine whether the following four statements are correct, and fill in **T** (for correct) or **F** (for incorrect) in the corresponding column of the table below .

(A) Patients with high levels of MHC-1 expression in cancer cells respond better to treatment with anti-CD47 antibodies.

(B) Patients with high MHC-1 expression in cancer cells respond better to treatment with activated cytotoxic T cells.

(C) Both CD47/SIRP γ and MHC-1/LILRB1 pathways can serve as targets for cancer immunotherapy.

(D) MHC-1 carries tumor antigens from inside the cell to the cell membrane, presenting them to helper T cells for recognition, thereby inducing adaptive responses.

Adaptive immunity

Narrative	(A)	(B)	(C)	(D)
T or F				

2. Clostridium tetani can infect the human body through wounds, releasing tetanus toxin to attack nerve cells and inhibit the neurotransmitter glucagon.

The release of glycine and γ -aminobutyric acid (GABA), and the reduction of glycine, leads to excessive stimulation of motor neurons.

Stimulation can lead to symptoms such as muscle stiffness. Recent studies in mice have revealed that nestin, which covers the surface of cells, plays a crucial role in this process.

(nidogen) is key to the entry of tetanus neurotoxin into the nervous system; the neurotoxin must first adhere to the surface of nerve cells.

Nestin then enters nerve cells via the surface of the cell and spreads its toxicity throughout the nervous system. Nestin is an extracellular matrix...

Extracellular matrix (ECM) proteins participate in various protein interactions to maintain tissue integrity; the nervous system is one such example.

One of them. Nestin covers the area around nerve cells, attracting the accumulation of tetanus neurotoxin; the terminals of motor neurons contain...

It contains a large amount of nestin, so even low amounts of the toxin can produce strong toxicity.

Determine whether the following four statements are correct, and fill in **T** (for correct) or **F** (for incorrect) in the corresponding column of the table below .

(A) Administering tetanus toxoid can prevent tetanus toxin from entering nerve cells, thus treating tetanus.

(B) Nestin can be used as a tool to deliver drugs into nerve cells, just as modified viruses can be used to transport drugs.

Viral therapy involving the delivery of target cells.

(C) The extracellular matrix is a barrier for the host to resist pathogen invasion. Clostridium tetani can degrade cells using toxin-binding proteins.

Entering the nervous system through the extracellular matrix

(D) The extracellular matrix is a highly dynamic network of macromolecules synthesized and secreted by animal cells outside the cell.

Composed of glycoproteins, the components of which can determine the properties of tissues; common components include collagen and fibronectin.

Proteins such as fibronectin, elastin, and laminin. Their functions include supporting cells.

Cells grow, differentiate, and adhere, and can serve as important conduits for transmitting information.

Narrative	(A)	(B)	(C)	(D)
T or F				

3. Many species in the genus ***Leishmania*** can infect humans, causing skin and visceral infections, respectively.

Leishmaniasis manifests in various forms, including sexually transmitted infections and mucosal infections, affecting approximately 12 million people worldwide. The eukaryotic mononuclear cells of Leishmania...

These are individual cells that can parasitize and reproduce within human host cells. Host defense mechanisms against Leishmania include the use of macrophages...

Phages produce reactive oxidants (ROS) and reactive nitrogen species (RNS) to eliminate infection, but many Leishmania species can still survive in the human body for more than 40 years.

Studies have found that Leishmania lacks catalase, but does possess peroxide reductase.

(peroxiredoxins), thiols, kinases, phosphatases, and ascorbic acid peroxidase

γγ(ascorbate peroxidase)γ

Determine whether the following four statements are correct, and fill in **T** (for correct) or **F** (for incorrect) in the corresponding column of the table below .

(A) The ability to evade the host's anti-leishmaniasis antibodies is key to the pathogen's ability to survive in the host for 40 years.

(B) The host's ability to resist Leishmaniasis infection, and the host's ability to produce interferon-γ (IFN-γ) and IFN-γ receptors.

Positive correlation

(C) After macrophages engulf pathogens, they form phagosomes. Leishmania uses these phagosomes for defense.

The creature evades the poisoning mechanism, survives and reproduces within the engulfing microorganism.

(D) Pattern recognition receptors (PRRs) on the surface of macrophages and Leishmania protozoa

Molecular action is an important mechanism for the host's adaptive immune response.

Narrative	(A)	(B)	(C)	(D)
T or F				

4. Scientists are attempting to treat cancer by strengthening the human immune system through various methods, a practice known as cancer immunotherapy. If a cancer patient's immune cells are unable to recognize and fight tumor cells, scientists have considered borrowing immune cells from healthy donors to fight cancer together. The research team first mixed tumor cells with gene mutations with immune cells from healthy donors to stimulate the activation of these immune cells. Then, targeted components from the activated immune cells were implanted into the cancer patient's immune cells, enabling the cancer patient's immune cells to successfully recognize cancer cells. T cells generally have the ability to recognize abnormal cells and destroy tumors, effectively distinguishing invaders through protein fragments displayed on the surface of other cells. Although cancer cells also express foreign protein fragments or neoantigens on their surface, they can evade T cell recognition and attack. Scientists collected samples of tumor cells from melanoma patients...

All exogenous protein fragments were analyzed, and the patient's T cells were then tested for recognition. The results showed that although the patient's cancer cells presented a large number of different exogenous protein fragments, the patient's T cells were still unable to recognize the abnormal protein fragments from the tumor cells. T cells from healthy donors can successfully identify many abnormal protein fragments that T cells from patients cannot.

Determine whether the following four statements are correct, and fill in **T** (for correct) or **F** (for incorrect) in the corresponding column of the table below .

- (A) Research results show that using immune cells from healthy donors to "outsource" cancer immunity is an unattainable dream in cancer immunotherapy.
- (B) If you want to use someone else's immune cells to fight cancer, you need to identify foreign protein fragments that can be recognized by T cells and identify cancer cells. Abnormal cell fragments and methods to find more effective ways to isolate T cells.
- (C) Experts extract bone marrow from healthy donors, separate stem cells and T cells, and inject the stem cells into patients with plasma cell leukemia. The T cells are first cultured in vitro and exposed to a protein called WT1, which often attaches to the surface of cancer cells, before being injected into the patient. This approach has proven effective in treating patients. This immunotherapy can cure... All cancer patients
- (D) Co-culturing melanocytic carcinoma cells and mouse T-cytotoxic cell precursors (CTL precursors; abbreviated as CTL-PS) in vitro allows for antigen recognition, but without a common stimulating signal, CTL-PS cannot proliferate and differentiate into effector T-cytotoxic cells. If the gene encoding the B7 ligand is transfected into melanocytic carcinoma cells, CTL-PS will differentiate into effector T-cytotoxic cells to kill cancer cells. This method of activating CTL-PS by providing a common stimulating signal (B7 ligand) through gene transfection falls under the category of cancer immunotherapy.

Narrative	(A)	(B)	(C)	(D)
T or F				

Part One

2018 Selection Camp Theory Student Code:

5. The cell membrane is mainly composed of lipids, proteins, and carbohydrates. The main components of lipids are phospholipids and cholesterol. Phospholipids can be further divided into...

Phosphoglycerides and sphingomyelin (SM). Phosphoglycerides include phosphatidylcholine (lecithin).

Phosphatidylcholine (PC), phosphatidylserine (PS), phosphatidylethanolamine (Cephalosporin)

SM contains phosphatidylethanolamine (PE) and phosphatidylinositol (PI). SM is present in nerve tissue.

It is abundant in the myelin sheath. SM on the cell membrane is hydrolyzed by sphingomyelinase (SMase) to produce ceramide, an important lipid in the extracellular matrix that tightly binds and maintains...

The integrity of the stratum corneum lipids forms a natural waterproof film, which is also an active ingredient in many emerging food and cosmetic products in recent years.

It can affect cell growth, division, differentiation, and apoptosis, thus delaying skin aging, maintaining youth, improving immunity, and fighting [diseases/promoting]...

With effects such as tumors, the annual output value of neurolamine products in the United States exceeds US\$10 billion.

Determine whether the following four statements are correct, and fill in **T** (for correct) or **F** (for incorrect) in the corresponding column of the table below .

(A) Ceramides are related to the structure of biological membranes and the composition of the extracellular matrix, and are also part of the innate immune defense against pathogen invasion.

A line of defense

(B) The lipid composition of the membrane lipid bilayer is asymmetrical; PC is mainly distributed in the inner layer facing the cytoplasm, while PS, PE, and...

PI is mainly distributed in the outer layer of the membrane.

(C) Myelin sheaths protect the axons of neurons and are extremely important for the transmission of neural information. A lack of myelin sheaths will damage them.

Nerve signals are transmitted more slowly at the damaged site, leading to poor muscle response and causing multiple sclerosis (MS).

Sclerosis)

(D) Genetic defects in SMase can cause SM to accumulate in the brain, liver, kidneys, and spleen, leading to delayed brain development and affecting intelligence.

Some children even die young, resulting in Niemann-Pick disease, an autosomal recessive genetic disorder.

Narrative	(A)	(B)	(C)	(D)
T or F				

6. Ribosomes are organelles in cells responsible for protein synthesis. Because cells need to rapidly adjust their responses to changes in the external environment,

And it undergoes cell division in a short period of time, so the total amount of mature ribosomal RNA in eukaryotic cells can reach [amount missing].

80% of the total. In eukaryotic chromosome sets, the number of ribosomal gene sets can sometimes reach 20,000. The nucleus of eukaryotes...

The glycosome gene is a repetitive gene arranged in a vertical line, head to tail. Each repetitive unit contains a small subunit and a large subunit.

The 5.8S and 8S genes contain coding regions separated by spacers called internal transcribed spacers (ITS). Large and small subunits are separated by external spacers, also known as intergenic spacers.

(Intergenic spacer, IGS), the 5.8S gene is located in the internal transcribed spacer region. Ribosomal genes, due to their high sequence...

Its highly conserved nature makes its gene sequences an important tool for constructing kinship relationships.

Determine whether the following four statements are correct, and fill in **T** (for correct) or **F** (for incorrect) in the corresponding column of the table below .

(A) Due to varying environmental pressures, the large and small subunit genes of the ribosome evolve at different rates in different regions.

Therefore, there are different levels of conservatism.

(B) Small subunit genes, due to their moderate size, are often used to construct kinship relationships.

(C) Large subunits contain many structural domains and evolve relatively rapidly, therefore they are not very conserved in terms of sequence.

It is not suitable for building kinship.

(D) The 18S and 28S genes also contain highly variable regions called expansion segments. In addition to...

Variations in the size of the 28S ribosome gene within the same species are due to differences in the number of extended segments and are not usually conclusive.

It is suggested to use it to construct kinship.

Narrative	(A)	(B)	(C)	(D)
T or F				

7. The hygiene hypothesis is a medical hypothesis that suggests that children's lack of contact with infectious agents and symbiotic microorganisms during childhood may lead to health problems.

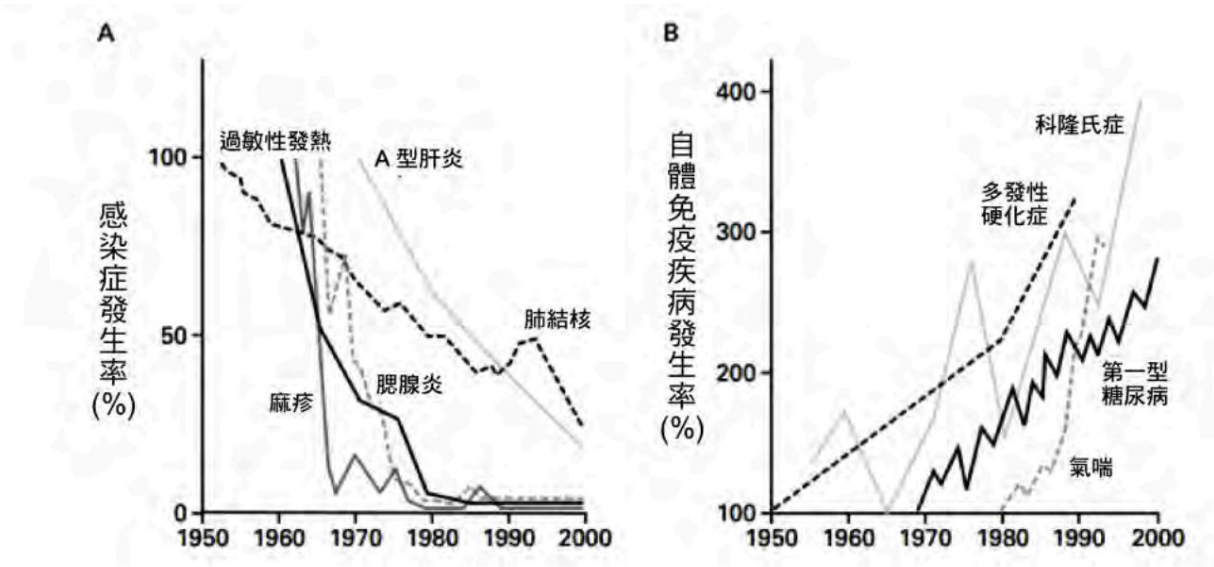
Foods (such as gut microbiota and probiotics) and parasites can restrict the normal development of the immune system, thereby increasing the risk of disease.

The likelihood of developing allergic diseases. National studies on the incidence of autoimmune diseases and acute lymphoblastic leukemia in adolescents have been developed.

The increase is believed to be related to the health hypothesis. An epidemiological statistic from the United States shows that from 1950...

Initially, perhaps due to vaccination and increased hygiene awareness, many infectious diseases such as measles, hepatitis, and tuberculosis became prevalent.

While the prevalence of certain diseases has decreased significantly, the prevalence of many immune disorders such as multiple sclerosis, type 1 diabetes, Crohn's disease, and allergic diseases such as asthma has increased dramatically. (See figure below)



Determine whether the following four statements are correct, and fill in T (for correct) or F (for incorrect) in the corresponding column of the table below .

- (A) Improved hygiene has induced autoimmune diseases.
- (B) The lack of exposure to various microbial environments leading to immune unresponsiveness is a contributing factor.
- Possible causes of autoimmune diseases
- (C) Vaccination with smallpox involves exposing children to a (potentially weakened) source of infection to reduce the incidence of the disease.
- (D) Evidence supporting the hygiene hypothesis: the interaction between the surface components of gut microbiota and the human immune system.

Narrative	(A)	(B)	(C)	(D)
T or F				

Part One

2018 Selection Camp Theory Student Code:

II. Short Answer Questions (25 **points total**)

1. Substances need to pass through the cell membrane to enter the cell. Please describe the following five ways in which molecules pass through the cell membrane.

(1) Water molecules **(2 points)**

(2) Oxygen molecules **(2 points)**

(3) Proton **(2 points)**

(4) Glucose molecules **(2 points)**

(5) Cholesterol molecule **(2 points)**

2. Briefly describe each of the following reactions (each should not exceed 100 words).

(1) Photorespiration **(5 points)**

(2) Aerobic respiration **(5 points)**

(3) Anaerobic respiration **(5 points)**

III. Multiple Choice Questions (15 points in total)

- _____ 1. What is the protein signaling molecule that can alter glucose uptake? And where is its receptor located? (3 points)
- (A) Insulin; various types of cells that use glucose as an energy source.
 - (B) Insulin; beta cells of the pancreas
 - (C) PDGF (platelet-derived growth factor); blood
 - (D) NGF (nerve growth factor); involved in reflexes
- _____ 2. In cellular signal transduction, what enzyme catalyzes the breakdown of phosphatidylinositol 4,5-bisphosphate (PIP₂) into inositol triphosphate (IP₃) and diacylglycerol? (3 points)
- (A) Phosphotransferase C
 - (B) Phospholipase C
 - (C) Phosphodiesterase C
 - (D) Lipokines
- _____ 3. Which of the following descriptions of the signaling molecules of steroid hormones is correct? (3 points)
- (A) It is produced only in specific locations within the body, but can still affect people from a great distance.
 - (B) Hydrophilic, therefore it cannot penetrate the cell membrane.
 - (C) Binds to receptors on the cell surface, triggering a chemical cascade reaction.
 - (D) Does not exist in human blood.
- _____ 4. Which of the following cell organelles is related to motor proteins? (3 points)
- (A) Smooth endoplasmic reticulum (ER)
 - (B) Vesicle
 - (C) Plasma membrane
 - (D) Chloroplast
- _____ 5. Lysosomes are specialized vesicles found in _____, containing digestive enzymes that break down food. Another related organelle is the vacuole, which exists in the _____ and contains enzymes but is often used for storage.
- Organelles. (3 points)
- (A) Animals; plants and fungi
 - (B) Plants; animals and fungi
 - (C) Plants and fungi; animals
 - (D) Animals and plants; fungi

IV. Short Answer Questions (10 **points total**)

As a scientist, you're attempting to clone a sheep using somatic cell nuclear transfer (SCNT). To achieve this, you first extract the nucleus from a mammary gland cell in sheep A and transfer it to an enucleated egg cell in sheep B. Next, you implant this enucleated egg cell into a pseudo-pregnant sheep, C, and name the newborn cloned sheep D. You discover that sheep D is born with the same genetic disease as sheep B.

Please explain why this phenomenon occurs and what factors might lead to this disease. (10 **points**)

V. Fill in the blanks (1.5 points **per blank** , **15 points in total**)

Ribosomes, formerly known as ribonucleoprotein bodies or nucleoprotein bodies, are organelles in cells, consisting of two ribosomes, one large and one small.

It is formed by the combination of subunits, and its main component is intertwined RNA (called "ribosomal RNA").

The nucleus contains rRNA (referred to as rRNA) and proteins (called ribosomal proteins, abbreviated as RP).

Glycosomes are the site of protein synthesis within cells. They can read the genetic information contained in the nucleotide sequence of messenger RNA and convert it into protein.

This information is converted into the sequence of amino acids in proteins to synthesize them. This process is used in prokaryotes and eukaryotes (two species of organisms on Earth with...).

Ribosomes are present in the cells of all major life forms (the former can be further divided into archaea and eubacteria).

Generally speaking, prokaryotic cells have only one type of ribosome, while eukaryotic cells have two types of ribosomes (ribosomes in mitochondria and ribosomes in chloroplasts).

(The ribosomes in the body are different from those in the cytoplasm). Ribosomes can be classified according to their sedimentation coefficient.

Based on the magnitude of the sedimentation coefficient, ribosomes are classified as follows: 55S ribosomes, 70S ribosomes, and 80S ribosomes, etc.

The table below compares the sedimentation coefficients of ribosomes in eukaryotic and prokaryotic cells. Please fill in the correct answer option (sedimentation coefficient).

Enter in the table:

Answer Options	(A)	(B)	(C)	(D)	(AND)	(F)	(G)
	80S	70S	60S	50S	40S	30S	28S
Answer Options	(H)	(I)	(J)	(K)	(L)	(M)	(N)
	23S	20S	18S	16S	15S	5.8S	5S

Classification	Ribosome size	Subunit		rRNA RP types	
		Large subunit	Small subunit		
prokaryotes					31
					21
eukaryotes (Cytoplasmic ribosomes)					49
					33

2018 International Biology Olympiad

National Team Selection Camp

Theoretical Assessment Test Questions

Part Two

Student Code



Precautions:

1. This section of the test consists of 6 pages (including the cover) and is worth 100 points. Please check for any missing pages.

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3. Please answer the questions on the test paper using a blue (black) pen, and please keep the test paper clean.

4. You may write your answers on the back of the questions, but please indicate the question number.

Part Two: (100 points)

The following are topics related to plant water physiology. Please answer the questions according to the descriptions for each topic.

I. Plant Water Potential: Botanists often use plant

water potential to predict its water content and other water physiology parameters, in order to understand the physiological functions within the plant.

And its mechanism of action. Applying this knowledge to agricultural technology and cultivation management allows for the control of water supply strategies and the improvement of crop production.

Potential and reduced production costs. The following diagram (a) shows the structure of a plant leaf, and (b) shows the water conduction pathway and resistance: (Total 25 points)

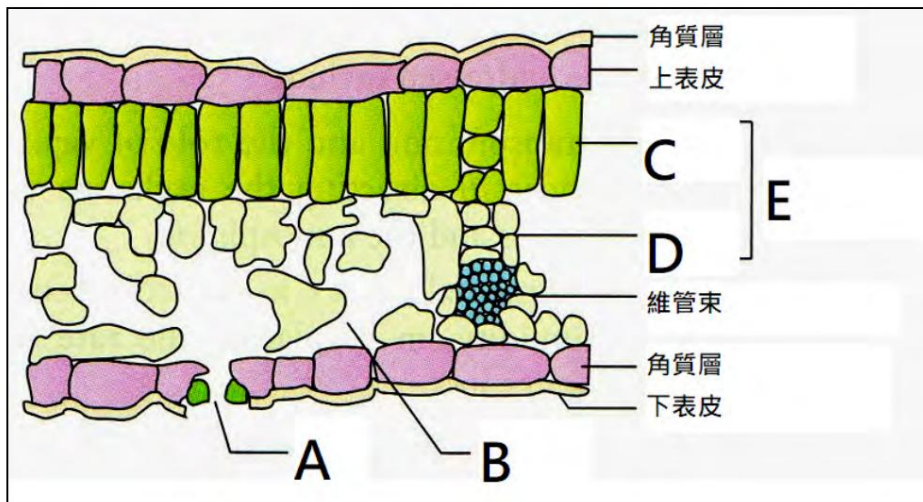


Figure (a) Plant leaf structure

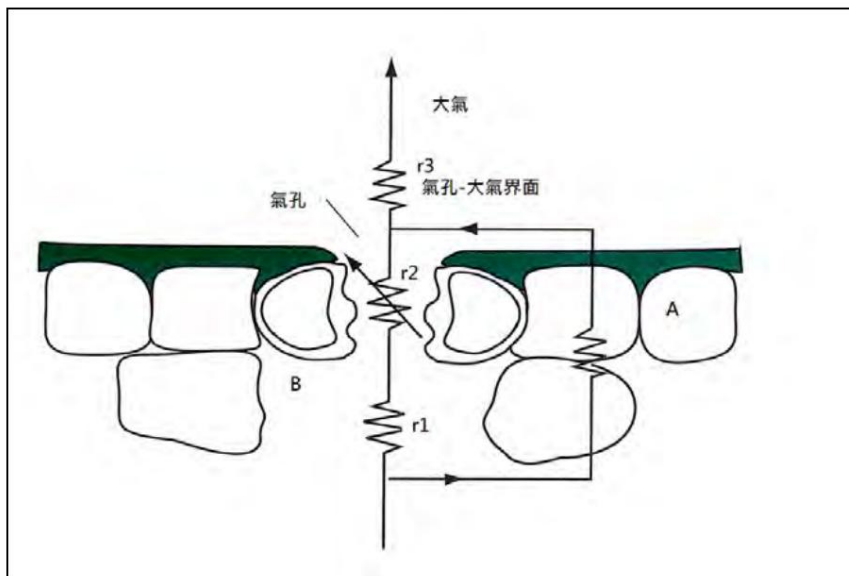


Figure (b) shows the water conduction pathway and conduction resistance (conduction resistance is represented by r).

The following is a set of questions related to water potential in plant leaves. Please answer the questions in the following set:

(1) Please fill in the structure of the plant leaf in Figure (a): A _____, B _____, C _____.

D _____, AND _____ (10 points)

Part Two

2018 Selection Camp Theory Student Code:

(2) In Figure (b), plant physiologists use the physical symbol r to represent the resistance to water conduction. Please explain what kind of water conduction resistance r_1 , r_2 , and r_3 represent respectively? At noon on a windless, sunny summer day, which of the three resistances is the greatest? Which is the next greatest? Please explain your reasoning (10 points). Answer:

(3) Following on from the previous question, comparing the gentle breeze and sunrise after rain in the early morning with the sunny midday in question (2), is the value of r_3 larger in the former environment (gentle breeze and sunrise after rain)? Please explain your reasoning (5 points). Answer:

II. Transport of photosynthetic products

In experiments conducted by botanists studying the transport of photosynthetic products to storage organs, it was discovered that water physiology significantly affects the transport capacity of photosynthetic products, and water potential can be used to explain and predict the transport capacity of substances. Based on this, please answer the following questions: (Total

25 points) (1) How can the transport mechanism of photosynthetic products in the vascular bundles of plant stems be explained using water potential? (10 points)

answer:

(2) How is the water state of the phloem in the vascular bundles expressed by the water potential formula? What are its units? (5 points)

answer:

(3) What is the pressure-flow hypothesis? How does this theory verify the transport of photosynthetic products to the storage tank?

Official? (10 points) Answer:

Part Two

2018 Selection Camp Theory Student Code:

The following are plant diversity and related topics. Please answer the questions according to the descriptions for each topic.

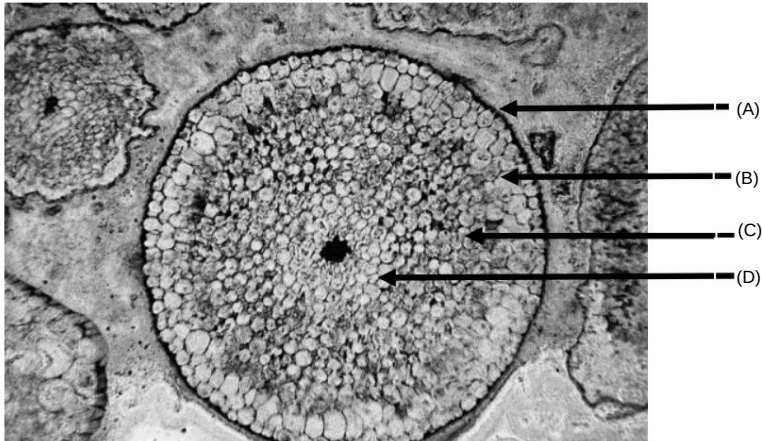
I. Plant Anatomical Characteristics The

external and internal structures and characteristics of plants form the basis of plant growth and physiological functions. Therefore, understanding plant characteristics can lead to further inferences.

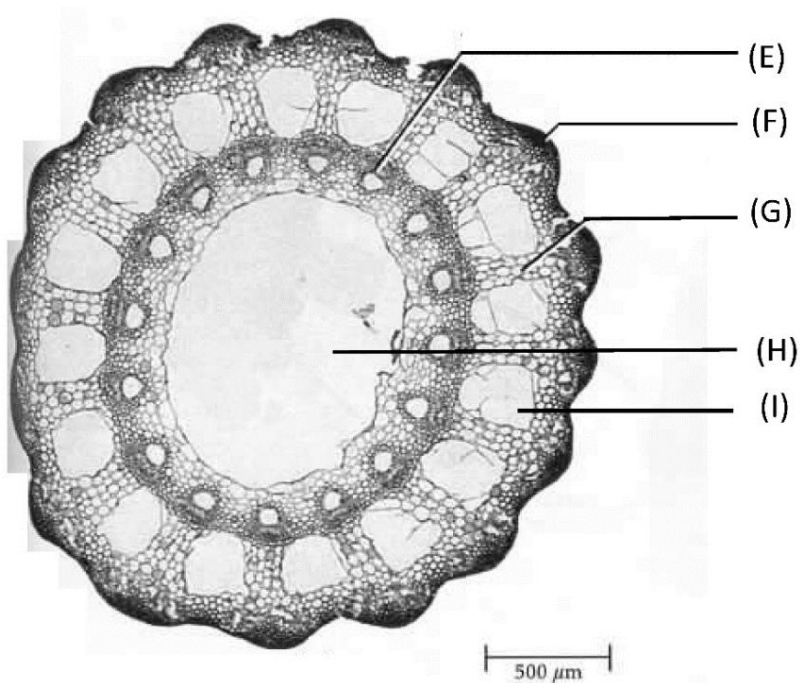
Discussing the relevant evolutionary history. The image below (a) shows a cross-section of the stem of the extinct *Rhynia* fern, and (b) shows a cross-section of the stem of *Equisetum*.

Cross section: (25 points total)

(a)



(b)



The following are sets of questions related to plant anatomy. Please answer the following questions: (2 points * 8 = 16 points)

(1) Please fill in the following structure: (A) _____ (B) _____ (C) _____

(D) _____ (AND) _____ (F) _____ (G) _____

(H) _____

(2) To which type do the vascular bundles of the stems of "*Lysimachia foetida*" and "*Equisetum hyemale*" belong? (2 points * 2 = 4 points)

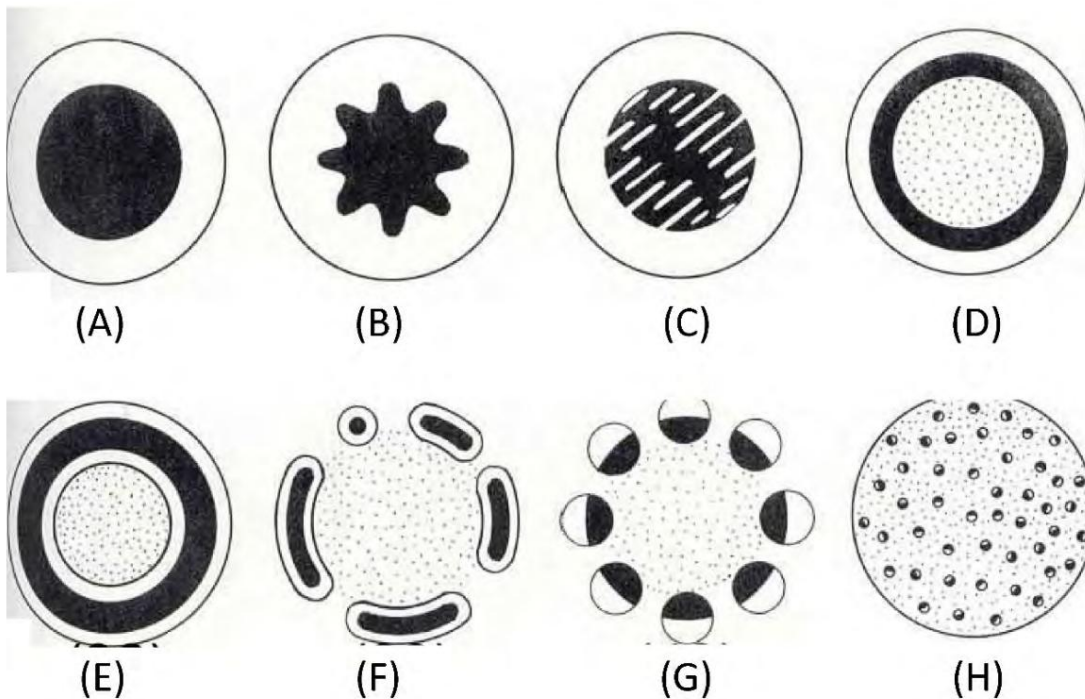
Please explain the differences between the two. (5 points)

Answer:

II. Characteristics of Plant Evolution

The structural forms of the stele in plants evolved into different types after they entered land, which is related to the evolutionary branching of terrestrial plants.

Based on the relevant connections, the following diagram shows the main types of central pillars. Answer the following questions based on the diagram: (Total 25 points)



(1) Fill in the corresponding code from the above diagram in the right column of the table below.

(1.5 points * 8 = 12 points) Atactostele	
True center post (Eustele)	
Plectostele (woven center column)	
Dictyostele	
Actinostele	
Amphiphloic siphonostele	
Haplostele (single center post)	
Ectophloic siphonostele	

Part Two

2018 Selection Camp Theory Student Code:

(2) Based on the structural characteristics of the eight different types of central stele in the figure above, nonvascular plants were selected as the outgroup.

(outgroup) Draw a phylogenetic tree for all vascular plant groups, and use it as shown in the figure above.

Variations in column structure are considered a feature, and at least one derived characteristic should be indicated on each clade of the phylogenetic tree. **(13 points)**

Theoretical Assessment Questions for the 2018 International

Biology Olympiad

National Team Selection **Camp**

Part Three

Student Code



Precautions:

1. This section of the test consists of 5 pages (including the cover) and is worth 100 points. Please check if there are any missing pages.
If any pages are missing, please inform the test administrator immediately.
2. When answering the questions, please check that the cover of the test paper matches your test number.

3. Please answer the questions on the test paper using a blue (black) pen, and please keep the test paper clean.
4. Please fill in the answers to questions 1, 2, and 3 on the question paper.
5. Please fill in the answers to questions 4 and 5 on the answer sheet.

Part Three: (100 points)

1. Please answer the following questions (2 points each, 16 points in total)

(1) In the testicles _____ (Cells) secrete _____ This substance can be affected _____

Regulates and assists in spermatogenesis.

(2) The cells in the ovary that secrete androgens are... _____

(3) In the ovary _____ Will utilize _____ Transform male hormones into female hormones

It assists in oogenesis.

(4) During embryonic development, in the testes _____ secretion _____ Suppressing female

Reproductive organ development

2. Carefully read the following description, select the key information and values, and then perform the calculation.

Generally speaking, oxygenated blood refers to blood with oxygenated hemoglobin when the oxygen partial pressure is 100 mmHg.

Blood with an oxygen saturation percentage γ 97.5% and dissolved oxygen of 0.3 mL/dL is considered oxygenated blood. Deoxygenated blood, on the other hand, is defined as blood with an oxygen partial pressure of 40 mmHg, an oxygen saturation percentage γ 75%, and dissolved oxygen of 0.3 mL/dL.

The oxygen content of blood is 0.12 mL/dL. For every 100 mL of blood flowing through surrounding tissues, the body consumes approximately 5 mL of oxygen. When heme is completely saturated with oxygen (each gram of heme binds 1.5 mL of oxygen),

If each liter of blood contains 200 grams of heme, what is the cardiac output (15 kg/min)?

How much oxygen can be delivered from the left ventricle to the aorta per liter? (The formula and calculation steps must be shown; otherwise...)

(No score will be given) **(12 points)**

Part Three

2018 Selection Camp Theory Student Code:

3. The diagram on the right shows a motor nerve receiving innervation from external nerves (A-E).

Intention, where nerves A, C, D, and E are the nerves released when stimulated.

The neurotransmitter is glutamate; released when the B nerve is stimulated.

The neurotransmitter is GABA.

(1) Fill in the blank: When nerves A through E are stimulated respectively (inadequate stimulation intensity)

When this nerve is induced to generate an action potential, if the electrical activity is recorded...

The five nerves (A-E) record changes in cell membrane voltage at the poles of motor nerve cell bodies.

The order of sexual influence from greatest to least is as follows (listed by ">" symbols)

(Indicate the order of size) (10 points)

answer: _____ > _____ > _____ > _____ > _____

(2) Multiple choice question: If stimulation is performed continuously within a short period of time (5 milliseconds)

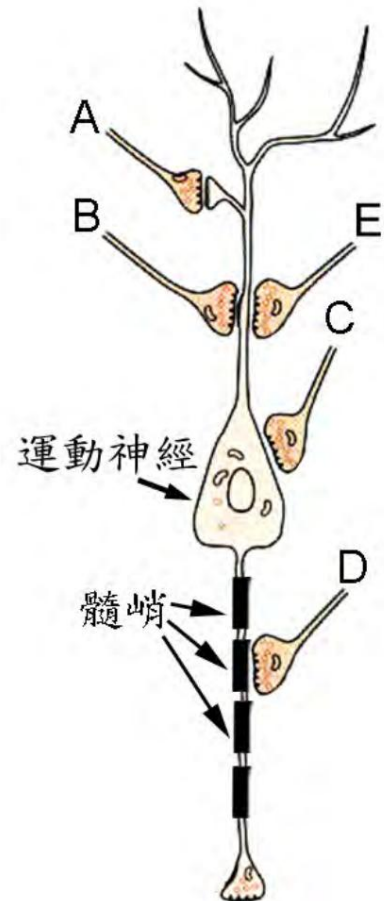
Given nerves (A, C, D, or E) and B, then the following combinations...

The middle γ nerve exerts the greatest influence on the membrane voltage of motor nerve cells. _____

What is the combination that produces the strongest effect? (2 points)

Why? (3 points)

answer: _____



(A) Stimulate A first, then stimulate B 5 milliseconds later. (B) Stimulate B

first, then stimulate A 5 milliseconds later. (C) Stimulate C first, then

stimulate B 5 milliseconds later. (D) Stimulate B first, then stimulate C 5

milliseconds later. (E) Stimulate E first, then stimulate B 5 milliseconds

later. (F) Stimulate B first, then stimulate E 5 milliseconds later. (G)

Stimulate D first, then stimulate B 5 milliseconds later. (H) Stimulate B first,

then stimulate D 5 milliseconds later.

(3) Question and Answer: When a motor nerve receives an external stimulus that reaches a threshold and generates an action potential, the action potential is transmitted to...

What changes occur at the nerve endings that lead to the release of neurotransmitters? (10 points)

answer:

4. Figure 1 (please turn to the next page) shows one of the recent hypotheses regarding the phylogenetic relationships of major animal groups.

Please answer the following questions based on this hypothesis. Please note that all answers can be supplemented with both text and diagrams to clearly express your views.

If we use the concept of parsimony to predict character evolution, how many times does the following feature independently gain and lose? If a feature exhibits polymorphism within a group, please also indicate this and explain why you cannot determine the number of times the feature appears and disappears. (30 points) (Please write your answer on the answer sheet.)

(1) Head

(2) Internal fertilization

(3) Filter feeding

(4) Floating larva (5) Aquatic

(6) Molts

during growth

5. The ecological phenomenon that "animals are larger when the ambient temperature is low and smaller when the temperature is high" is known in academic circles as "Bergmann's rule." When defining this phenomenon, Bergmann himself limited its application to closely related species within the same genus. However, later scholars often extend Bergmann's rule to explain size differences between different populations within the same species. (Please write your answer on the answer sheet.)

In 2011, biologists Jennifer A. Sherida and David Bickford from the National University of Singapore discussed the relationship between animal body size and geographical environment, proposing the startling argument that "the shrinking size of animals reflects the reality of global climate change." Subsequently, American and Dutch geologists and archaeologists discovered that the body size of the ancient horse (*Arenahippus*) and the ancient raven deer (*Diacodexis*) was indeed at its smallest during periods of highest average temperature.

However, a research team from the University of Florida compared data from bird and mammal species across 30 different geographic regions, conducting a correlation analysis between body weight and geographic environmental factors. In a total of 952 bird and mammal species and 273,901 specimen records, they found no significant correlation between individual body weight and ambient temperature for most species (79%). Only 14% of species conformed to the Borgenman theorem, but 7% exhibited a phenomenon completely contrary to the theorem's predictions—that is, the closer the habitat was to the equator, the larger the body size.

Therefore, how do you explain why the correlation between body size and temperature in most animals does not conform to Borgenman's theorem? If this theorem were applied to poikilothermic and aquatic animals, what results do you predict might occur? (17 points)

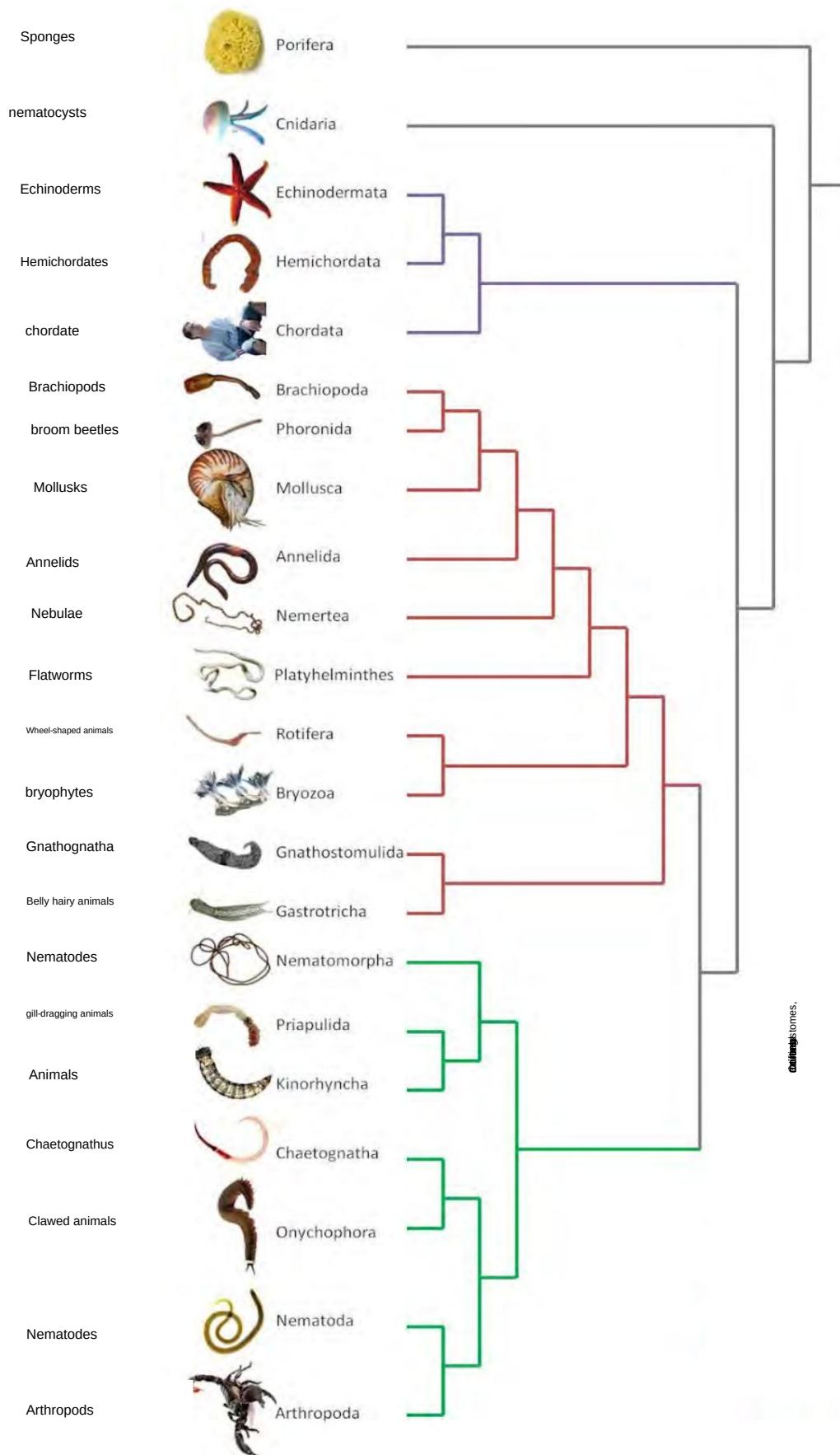


Figure 1

2018 International Biology Olympiad

National Team Selection Camp

Theoretical Assessment Test Questions

Part Four

Student Code



Precautions:

1. This section of the exam consists of 4 pages (including the cover) and is worth 100 points. Please check for any missing pages.

Please inform the test administrator immediately if there are any missing pages.

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4. Please write your answers on the answer sheet and indicate the question number.

Part Four: (100 points)

1. During avian embryonic development, the endoderm epithelial cells (EEC) absorb nutrients from the yolk and distribute them to the embryo for its development.

A gene (SOAT1) expresses an enzyme that adds a fatty acid to cholesterol to form cholesterol esters, according to research.

This gene plays an important role in embryonic development, especially in enhancing the utilization of yolk fat.

Because the expression of this gene significantly affects the efficiency with which the embryo utilizes fat in the yolk, it also alters the embryo's developmental rate.

Therefore, understanding the regulatory mechanisms of gene expression is crucial. Adding cAMP-promoting drugs to EEC cell culture medium...

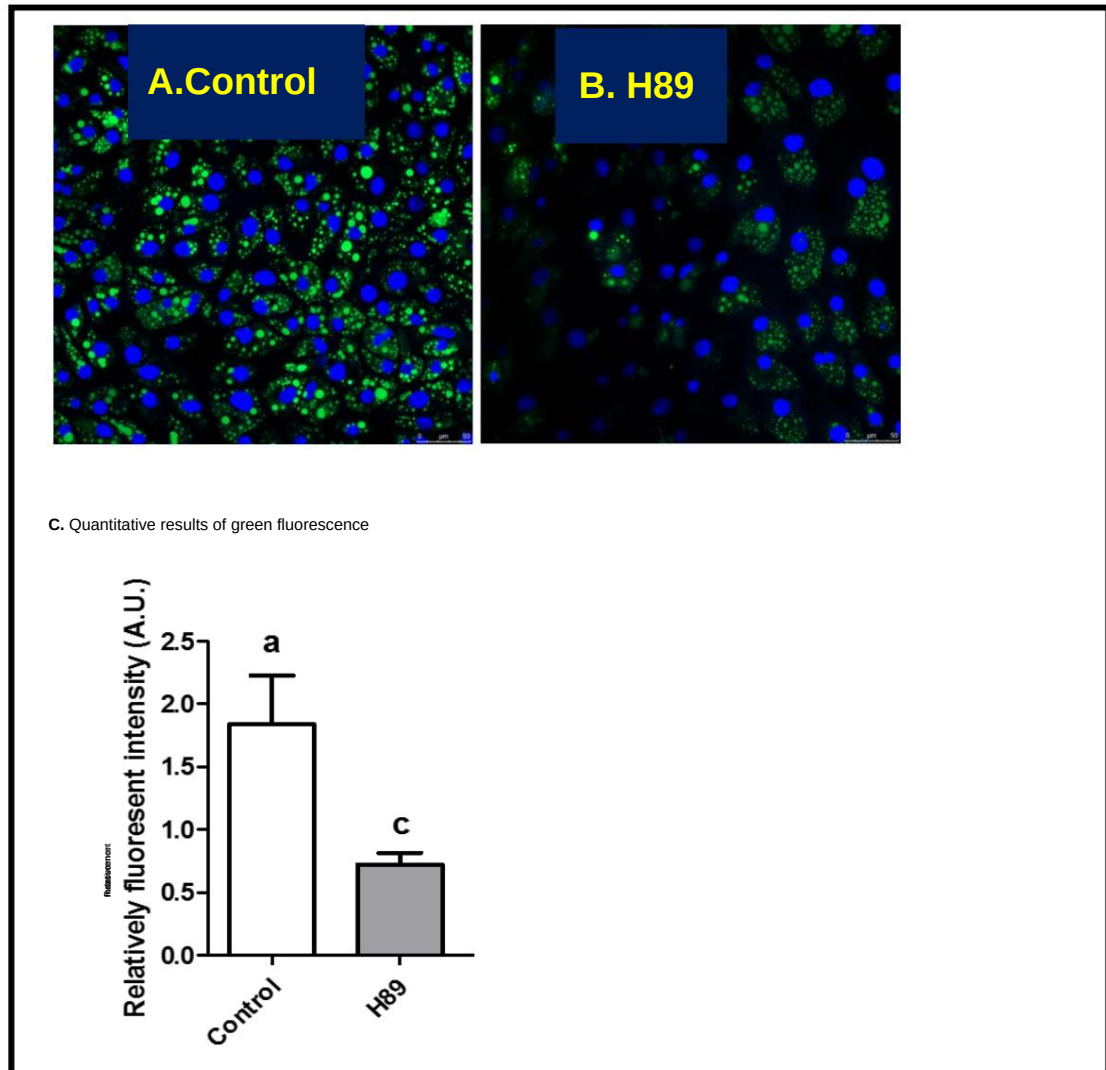
The agent was found to enhance the expression of this gene and also enhance the function of the cAMP target gene PKA.

1. Given that PKA promotes the phosphorylation of its target proteins, how can the functional performance of PKA be analyzed? **(10 points)**

2. H89 is an inhibitor of PKA; the addition of H89 significantly reduces the production of cholesterol esters by SOAT1 (Figure).

(i) Furthermore, research has also found that PKA promotes the expression of the SOAT1 gene. What is the application of H89 to EEC cells?

How is SOAT1 mRNA expression affected in the culture medium? **(10 points)**



(Figure 1)

(Figure 1) H89 is an inhibitor of PKA. After adding H89, the cholesterol esters produced by SOAT1 are significantly reduced (Figure A above).

(And the green fluorescent oil droplets in B), blue represents the staining results of cell nuclei. In Figure C, a and c represent treatments A and B, respectively.

There are significant differences between the two groups.

(Please fill in your answers on the answer sheet)

II. Following on from the previous question, if scientists add a reporter gene to the full-length promoter of the SOAT1 gene, they can determine this promoter...

The mechanism by which the motor is regulated.

1. If a cAMP inhibitor is added, what will be the observed state of expression of this reporter gene? (5 **points**)
2. If a drug that antagonizes PKA is added, what will be the observed state of expression of this reporter gene? (5 **points**)
3. If a drug that promotes PKA is added, what will be the observed state of expression of this gene? (10 **points**)
4. If another endorphin also promotes PKA function, how would the addition of this endorphin affect this reporter gene?

Performance? (10 **points**)

(Please fill in your answers on the answer sheet)

III. The 5' untranslated region (5'UTR) of some prokaryotic genes' mRNA can bind to specific metabolites (Met) and form different secondary structures to regulate the transcription or translation of mRNA. Because it can regulate gene expression, it is called a riboswitch. By linking this 5'UTR sequence to a luciferase reporter gene and constructing this recombinant reporter gene efflastosome, it can be introduced into cells. Analysis of the luciferase activity exhibited in the cells reveals the effect of the riboswitch on reporter gene expression. The experiment can also detect the luciferase activity of the reporter gene under the presence of metabolites (Met) to understand how the metabolites affect gene expression. Figure (a) below shows the luciferase activity units exhibited when the 5'UTRs of gene A and gene B's mRNA were ligated to a luciferase gene, with or without the addition of metabolites (Met). Figure (b) shows the effect of adding the transcription terminator inhibitor BCM on the 5'UTR riboswitching activity of genes A and B. Based on the above experimental results, please answer the following questions.

(A) Compare the effects of metabolites on the expression of genes A and B, and infer the influence of metabolites on the expression of the two genes.

The possible level is at the transcriptional or translational level. (25 points)

(B) Explain and compare how the 5'UTR riboswitch in genes A and B is regulated by metabolites. (25 points)

Figure (a) Effects of metabolites on cold photoenzyme activity

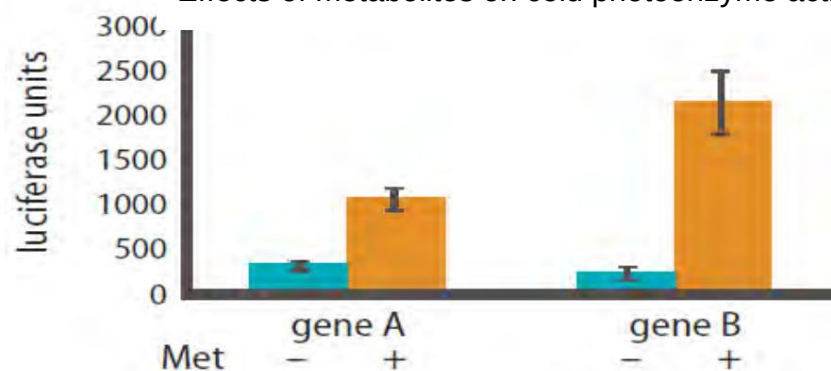
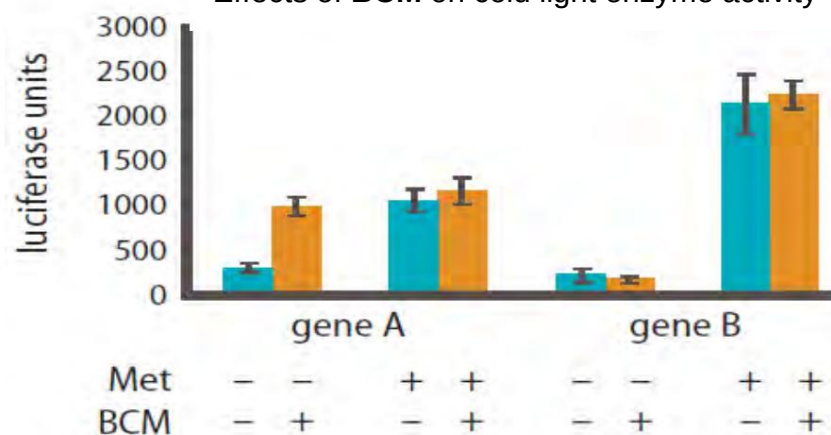


Figure (b) Effects of **BCM** on cold light enzyme activity



(Please fill in your answers on the answer sheet)

Theoretical Assessment Questions for the 2018 International

Biology Olympiad

National Team Selection **Camp**

Part Five

Student Code



Precautions:

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4. You may write your answers on the back of the questions, but please indicate the question number.

Part Five: (100 points)

I. Multiple Choice Questions (Single choice, 5 points each , 20 points in total)

_____ 1. In accordance with the 13th Sustainable Development Goal set by the United Nations in September 2015.

The greatest threat to human development today is (SDGS).

- (A) Climate Change
- (B) Pollution emissions
- (C) Melting of the polar ice caps
- (D) Population growth issues

_____ 2. To address climate change, countries adopted the "Paris Agreement" at the COP21 meeting in Paris on December 12, 2015.

The agreement stipulated that all countries should strive to limit the rise in global temperatures to a certain degree Celsius.

- (A) 1° Celsius (B) 2°
- Celsius (C) 5° Celsius
- (D) 6° Celsius

_____ 3. The main factors leading to species extinction in the last 500 years are:

- (A) Natural disasters (floods, fires, earthquakes, etc.)
- (B) Human activities (C)
- Large-scale deforestation of tropical rainforests
- (D) Climate change (e.g., rising temperatures)

_____ 4. During archaeological excavations, a group of leaf fossils was discovered, most of which had smooth edges, with only a few having serrated edges.

The environmental conditions at the time of deposition are:

- (A) The depositional area has an arid climate.
- (B) The local rainfall is sufficient.
- (C) The local climate is warm.
- (D) The local climate is cold.

II. Questions and Answers

(i) Regardless of the past hundred years (1873-1995) or the more recent fifty years (since the 1950s),

The most significant trend in East Asian climate change is the weakening of the winter monsoon. This is reflected in the intensity of the northeast monsoon and the strength of the Siberian High.

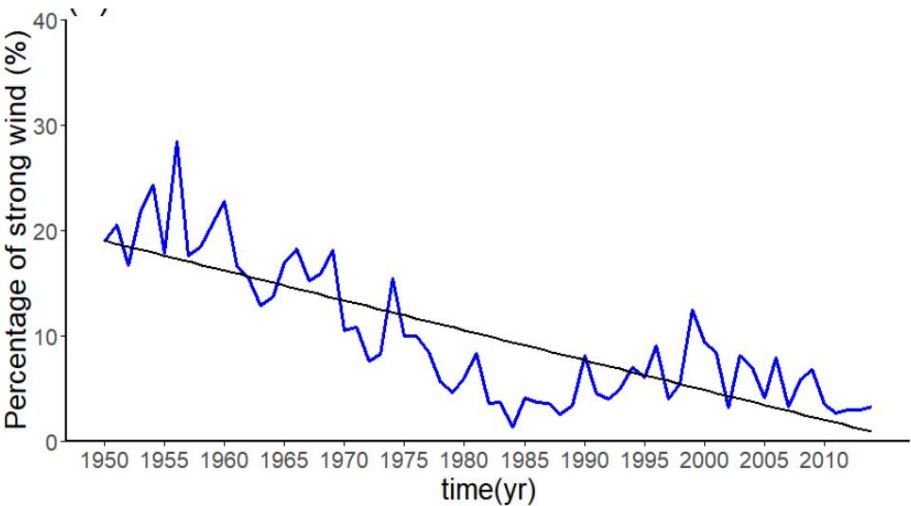
The temperature is positively correlated with the wind speed; when the northeast monsoon is strong, the temperature drops significantly and strong winds can occur in some areas.

The cold waves brought by the southward movement of the northeast monsoon caused extreme low-temperature events in various regions. (Figure 1) shows the data from the Dawu weather station from 1950 to 2013.

The percentage of strong northeast monsoon winds, while fluctuating in intensity, shows an overall downward trend. On the other hand...

Regarding temperature, due to global warming, Taiwan's temperature has risen by approximately 1.4 degrees Celsius over the past 100 years, and the increase...

Wen's pace has been the fastest in the last 30 years.



(Figure 1) Percentage of strong northeast monsoon winds at Dawu station from 1950 to 2013.

1. The impacts of climate change on the species composition and structure of Taiwan's forests

The Nanrenshan Forest Dynamics Sample Area (Permanent Sample Area) was established in 1990. All woody plants within the sample area with a diameter ≥ 1 cm are included.

All plants were measured, identified, their distribution maps were drawn, and they were each labeled with a number. The dynamic sample area included the area located on the mountaintop in the northeast...

The 3-hectare sample area along the Lanren Creek, influenced by the monsoon, and the 2.1-hectare sample area along the Nanren Mountain Creek, located in a secluded part of the creek valley, each sample area...

Three follow-up checks were conducted to understand the new growth, mortality, and growth of each plant in the sample area. After data processing and analysis, each...

The overall results of the re-examination are shown in Table 1 (Lanrenxi Sample Area) and Table 2 (Nanrenshan Sample Area), including the year, number of species, etc.

Plant density, tree cross-sectional area (biomass represented by the sum of trunk cross-sectional areas per hectare), and Simpson's dissimilarity.

Index (the probability that two randomly sampled individuals belong to different species, %), the larger the value, the higher the degree of discrepancy.

Table 1. Results of four surveys conducted in the Lanrenxi sample area.

Number of species and plant density (hectares) in the year of survey	Tree cross-sectional area (square meters/hectare)	Simpson's Dissimilarity		
1991	109	14,956	42.31	24.37
1997	106	14,644	47.47	23.72
2005	104	12,723	45.31	23.52
2013	100	11,911	47.14	22.78

Table 2. Results of four surveys conducted in the Nanrenshan Valley sample area.

Number of species and plant density	(hectares) in the year of survey	Tree cross-sectional area (square meters/hectare)	Simpson's Dissimilarity	
1993	106	3,289	38.67	14.33
2000	103	3,426	42.82	14.06
2008	104	3,758	42.20	15.43
2013	105	3,620	42.64	14.97

Compare the sample areas of the Lanren Creek (Table 1) and the Nanren Mountain Valley (Table 2), and simultaneously compare the sample areas of different years within the same forest (specific years).

Based on the results of the first and fourth surveys (excluding the first and fourth surveys), answer the following questions:

- (1) Based on the 2013 survey results, the most significant difference in forest characteristics between the Lanrenxi sample area and the Nanrenshan Valley sample area was:

Why? And speculate on the reasons. (14 points)

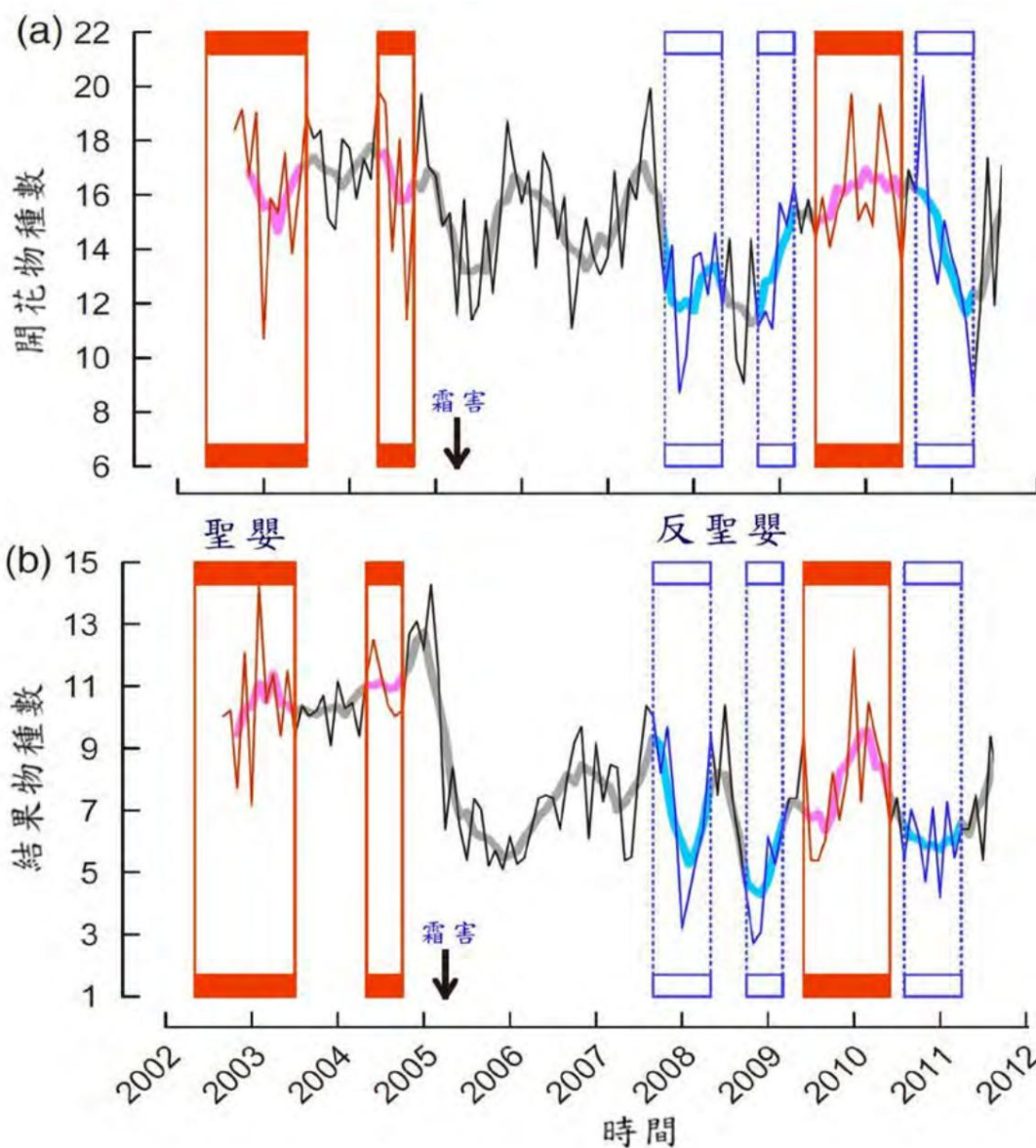
- (2) Compare the plant density and tree cross-section of the first (1991) and fourth (2013) surveys of the Lanrenxi sample area.

Based on this, what are some possible mechanisms by which climate change has led to this difference? (8 points)

2. The impacts of climate change on forest phenology in Taiwan

Eighty-seven collection nets were set up in a 25-hectare forest dynamics sample area in Fushan. For ten years, weekly monitoring of phenological phenomena such as flower and fruit drop was conducted within the sample area. Data analysis revealed that periodic climate variations (such as El Niño or La Niña) and extreme weather events (such as the frost damage in March 2005) significantly impacted flowering and fruiting in the Fushan area (Figure 2). El Niño and La Niña refer to abnormal sea surface temperature changes in the eastern-central equatorial Pacific Ocean (between the International Date Line and 120°W).

When El Niño or La Niña occurs, it causes climate anomalies in many regions, thereby affecting the local ecosystem. In East and Southeast Asia, during El Niño, the northeast monsoon weakens, reducing cold air from the north and resulting in warmer winters. In the Fushan area, the average monthly temperature rises, sunshine hours increase, and rainfall decreases. During La Niña, the northeast monsoon strengthens, leading to colder and wetter winters.



(Figure 2) The impact of El Niño and Anti-El Niño phenomena on the phenology of forest plants in Fushan (red box represents the El Niño period, blue box represents the Anti-El Niño period). The thin, wavy lines represent the number of flowering species (a) and fruiting species (b); the thick lines are trend lines, purple during the El Niño period, cyan during the Anti-El Niño period, and gray during other periods.

Compare the number of flowering species (Fig. 2a) and fruiting species (Fig. 2b) in Fushan Forest in different years, and answer the following questions:

(1) What is the impact of El Niño and Antineoplastic phenomena on the number of flowering species in Fushan Forest? (3 points)

(2) What is the impact of El Niño and Antineoplastic phenomena on the number of fruiting species in Fushan Forest? (3 points)

(3) What was the impact of the extreme weather event (the frost damage in March 2005) on the number of flowering species in Fushan Forest? (3 points)

(4) What was the impact of the extreme weather event (the frost damage in March 2005) on the number of fruit-bearing species in Fushan forest? (3 points)

(5) The annual recovery of the number of flowering and fruiting species after the extreme weather event (frost damage in March 2005) is as follows:

What are the possible reasons for this significant difference? (6 points)

(ii) The invasion of alien species can have a significant impact on many ecosystems around the world, including the potential to damage local native species.

The extinction of species.

1. Many invasive species can also be found in Taiwan's wild environment, including fish, amphibians, reptiles, birds, and mammals.

For each taxa, list one invasive species that can be observed in the wild, and that is not a domesticated animal or fowl (including cats and dogs).

Biology. (10 points)

2. When an invasive species invades, it may or may not compete with native species at the same food chain level: (I) Once it invades the area or habitat where the native species survives, the invasive species may directly displace the native species, leading to the disappearance of the native species' local population; (II) However, since invasive species often appear in environments with severe human disturbance, it is difficult for them to invade habitats with intact ecosystems; on the contrary, many native species appear in habitats with intact ecosystems, but cannot adapt to these environments with severe human disturbance. Therefore, after the invasion of an invasive species, native species may still coexist, that is, invasive species are distributed in environments with severe human disturbance, while native species survive in habitats with intact ecosystems.

(1) Suppose an invasive bird species A invades the distribution area of a native bird species B, which is the same size and food chain level as the native bird species B. Based on the above description, the following scenarios are possible: (I) bird species A competes with bird species B, or (II) bird species A and bird species B coexist. Assuming that the distribution of bird species A and bird species B can be accurately determined through direct observation, briefly describe what kind of species distribution phenomena would support the two hypotheses mentioned above (i.e., how to distinguish which of the two hypotheses is true)? (10 points)

(2) Briefly describe how to use manipulative experiments (e.g., removing birds) to further confirm either hypothesis (I) or (II).

Which one is correct? (10 **points**)

(3) If the native bird species B is scarce and the wildlife management unit wants to increase the number of bird species B,

Once it has been determined which hypothesis (I) or (II) is correct, how should wildlife management authorities respond to hypothesis (I)?

Or (II) to increase the number of bird species B? (10 **points**)